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# TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

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Paper under double-blind review

## ABSTRACT

Model merging seeks to consolidate multiple fine-tuned models into a shared weight representation that can serve the collection of source tasks. Although prior work shows that merged weights can often approach the performance of task-specialized models, the low-rank setting remains less understood, including models represented through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). We show that applying conventional merging methods directly to low-rank weights can cause severe degradation, indicating that conclusions from full-rank merging do not automatically transfer to compressed representations. Motivated by this behavior, we broaden the merging objective from producing one collapsed model to constructing a compact basis (e.g., an equivalent of holding two or more models) from which task-specific models can be recovered by linear combination. This view treats merging as the construction of a reconstruction-capable model space rather than as a single endpoint. It also enables the system to “revert” to an individual model when a task calls for specialization, reflecting the practical constraint that a single merged model need not dominate every task-specific model. We introduce Reversible Model Merging (RMM), an efficient, data-free, and flexible method with a closed-form solution for selecting basis weights and task-specific linear coefficients. Across the datasets, model scales, and low-rank compression regimes studied here, RMM consistently outperforms existing merging approaches and preserves low-rank model performance by a substantial margin.

## 1 INTRODUCTION

Parameter-efficient fine-tuning methods, such as Low-Rank Adapters (LoRA) (Hu et al., 2021), are now a standard way to adapt large language models across downstream tasks without paying the full cost of retraining every parameter. In parallel, compression methods such as post-training singular value decomposition (SVD) truncation provide another route to efficient storage and inference for task-adapted models (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). These developments make model merging increasingly relevant: rather than managing many independently adapted models, one can attempt to consolidate their task knowledge into a shared representation. Existing merging methods, from parameter interpolation to optimization-based procedures, show that merged models can often perform competitively across their source tasks without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025).

A central limitation remains, however: merging methods generally trade specialization for consolidation, and even strong strategies often fall short of the original individual fine-tuned models. This gap is especially relevant for low-rank model families, including LoRA adapters and SVD-truncated updates, because their compressed parameterization leaves less room to absorb conflicts introduced by merging. A single merged parameter vector can serve multiple tasks, but it also forces distinct task-specific adaptations into one shared space. In the low-rank setting, that collapse can amplify task interference and weaken the specialized directions learned by each model (Gargiulo et al., 2025).

Low-rank adapters and SVD-truncated updates are compact in isolation, yet large collections of them can still create a deployment bottleneck. Multi-task, federated, and continual learning work-

flows may accumulate many task-specific low-rank modules, each with its own storage and routing requirements. Keeping every adapter separately weakens the system-level savings that low-rank parameterization is intended to provide and complicates model sharing or deployment. This motivates merging strategies that operate directly on low-rank representations, not only to reduce redundant storage but also to support efficient management and transfer of compressed task adaptations across application settings.

We begin by examining whether existing merging techniques retain their behavior when the inputs are low-rank compressed models. Our analysis focuses on deltas, the weight differences between each fine-tuned model and a shared pre-trained initialization. Conventional methods combine these deltas into one parameter set, implicitly relying on sufficient linear compatibility among task updates. In our experiments, directly merging low-rank deltas produces catastrophic performance degradation. The failure is consistent with two interacting constraints: low-rank representations have limited expressive capacity, and independently optimized task subspaces can be misaligned, producing severe interference when they are forced into a single update.

This observation suggests a broader question about the scope of model merging: *What if the goal of merging were not to produce a single final model? What if we could relax the problem to allow retaining more than one model? What if we could preserve the ability to reconstruct the original models on demand?* Rather than collapsing multiple low-rank adapters into one set of weights, we maintain a compact collection of weights that acts as a shared basis from which each original model is recovered through linear combination. This reframing expands the design space of merging: instead of compressing all task information into a single model, it preserves complementary representational directions across a small number of basis models and enables post-merging “reversion” to the original adapters. The resulting perspective treats merging as construction of a reconstruction-capable model space, while retaining the same low-rank tasks and models considered in the empirical study.

Building on this reconstruction-driven view, we develop a theoretical framework for choosing a basis that minimizes reconstruction error. The resulting method, Reversible Model Merging (RMM), casts merging as a basis-selection problem: the objective is to find a compact set of task vectors that supports accurate reconstruction of the original task adaptations. RMM admits a closed-form solution that jointly identifies the basis task vectors and the task-specific coefficients used for linear recovery. We validate RMM across the datasets, compression choices, and model configurations included in this work, where it consistently surpasses conventional merging strategies by a significant margin. These results indicate that reversible merging can preserve the efficiency benefits of low-rank parameterization while giving practitioners a tunable storage-performance trade-off through the number of retained basis components.

## 2 RELATED WORK

Model merging covers a family of approaches for consolidating knowledge from multiple fine-tuned models, and this family is relevant across both full-rank and compressed adaptation settings. Existing methods can be viewed broadly as data-free parameter-space manipulations or as optimization-based approaches that use data or auxiliary objectives. We review representative techniques from both categories to position the reversible formulation developed here:

**Task Arithmetic:** Task arithmetic (Ilharco et al., 2023) is a foundational parameter-space view of merging. It defines a delta as the difference between the parameters of a fine-tuned model and its pre-trained initialization, then treats these deltas as modular building blocks. Such updates can be added to a base model to introduce abilities, subtracted to suppress behaviors, or scaled and combined to blend capabilities. This linear framework provides a general template for composing model behavior directly in weight space.

**DARE:** While simple delta addition can be effective, it often fails when merged models contain conflicting or overlapping parameter updates. To address this, Yu et al. (2024) introduced DARE (Drop and Rescale), a technique built on aggressive randomized pruning. The method operates by discarding the vast majority of parameter changes—up to 90-99%—on the premise that most updates are not essential and may even interfere with knowledge from other models. The remaining updates

are then amplified to preserve the overall scale of the modification, preserving core task-specific modifications while minimizing interference.

**TIES:** In contrast to DARE’s randomized pruning, TIES-merging introduces a deterministic framework for mitigating parameter conflicts (Yadav et al., 2023). It operates in three phases—Trim, Elect Sign, and Merge—each designed to systematically resolve interference across models. First, only the parameters with the largest magnitudes are retained, ensuring that the most influential updates from fine-tuning remain. Next, for each parameter position, the method evaluates the sign of updates across models and enforces agreement by discarding contributions that oppose the majority sign. Finally, the sign-consistent parameters are averaged to form the merged representation. This consensus-driven approach reduces destructive interference and yields a more stable merged model. Building on the same motivation of reducing interference as in DARE and TIES, EMR-Merging (Elect, Mask and Rescale) proposes electing a unified model and attaching task-specific binary masks and rescalers from each individual model to shift the unified model towards each task (Huang et al., 2024).

**Other Merging Methods:** Beyond data-free parameter manipulations, another class of merging methods uses optimization procedures or small auxiliary datasets. Model soups (Wortsman et al., 2022) tune interpolation weights by minimizing validation loss, steering the merged solution toward better generalization. Fisher-weighted averaging (Matena & Raffel, 2022) incorporates second-order information by assigning greater weight to parameters estimated to matter more for task performance. Other methods formulate merging directly as an optimization problem: RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) search for task-dependent coefficients using objectives based on prediction discrepancy and unlabeled prediction entropy, respectively. A recent approach minimizes the loss difference between a merged model and individual fine-tuned models through a first-order Taylor surrogate (Wei et al., 2025). These methods broaden the achievable performance of merging, but their reliance on labeled or unlabeled data, extra optimization, or both can limit their use when models must be consolidated without retraining or data access.

### 3 PRELIMINARIES

We consider the problem of consolidating a collection of fine-tuned models into a single unified model via model merging. Let  $\{\theta_1, \dots, \theta_n\}$  denote  $n$  models fine-tuned on  $n$  distinct tasks. A merging algorithm  $\mathcal{M}$  combines these models into a single set of weights  $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$ . Under this formulation, the desired procedure  $\mathcal{M}$  maximizes average validation performance over the task collection:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where  $\mathcal{D}_i$  denotes a held-out validation set for task  $i$ , and  $\text{Perf}$  denotes any suitable performance metric (e.g., accuracy or F1 score). It is standard to assume that all fine-tuned models originate from the same pretrained initialization  $\theta_{\text{pre}}$ . For any layer  $l$ , the update (or delta) of model  $i$  relative to the base model is defined as  $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$ , where  $\Delta_i^l \in \mathbb{R}^{m \times d}$  denotes the weight difference for a layer of dimensions  $m \times d$ .

#### 3.1 LOW-RANK COMPRESSED MODELS

Many deployment pipelines compress fine-tuned updates into low-rank form to reduce storage, training, and inference costs. The setting studied in this paper covers two widely used ways of obtaining such low-rank models:

**LoRA fine-tuning:** During adaptation, each layer receives trainable low-rank matrices  $A \in \mathbb{R}^{m \times r}$  and  $B \in \mathbb{R}^{r \times d}$ , where  $r \ll \min(m, d)$ . The pretrained weights remain frozen, so task learning is captured through  $A$  and  $B$ .

**Post-training SVD truncation (PT-SVD):** After standard fine-tuning, the update  $\Delta_i^l$  is decomposed by SVD and truncated to the top- $r$  singular components. This yields the low-rank approximation  $\hat{\Delta}_i^l = A_i^l B_i^l$ , with  $A_i^l \in \mathbb{R}^{m \times r}$  and  $B_i^l \in \mathbb{R}^{r \times d}$  where  $r \ll \min(m, d)$ .

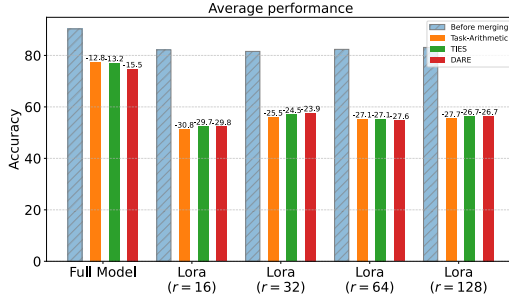


Figure 1: Average task performance before and after merging across the evaluated methods. The comparison covers fully fine-tuned models and LoRA-adapted models at ranks  $r \in \{16, 32, 64, 128\}$ .

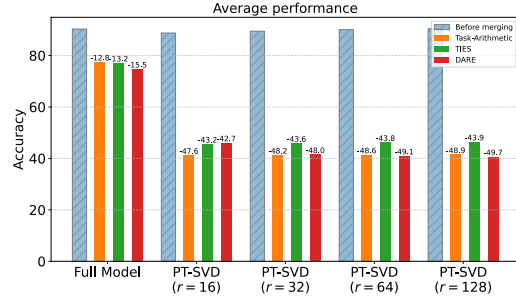


Figure 2: Average task performance before and after merging across the evaluated methods. The comparison covers fully fine-tuned models and PT-SVD-compressed models at ranks  $r \in \{16, 32, 64, 128\}$ .

Both mechanisms give the same abstract representation for layer  $l$  of model  $i$ :  $\hat{\Delta}_i^l = A_i^l B_i^l$ , where  $A_i^l \in \mathbb{R}^{m \times r}$  and  $B_i^l \in \mathbb{R}^{r \times d}$ , with  $r$  denoting the compression rank. The merging problem therefore becomes one of consolidating low-rank updates across tasks. In the conventional single-model formulation, the merged model integrates the compressed deltas  $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$  across all tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where  $L$  denotes the total number of layers. Throughout this work, we study this low-rank compressed setting while keeping the underlying tasks, models, and evaluation protocols fixed. The next section examines how conventional merging strategies behave when applied within this setting.

#### 4 MOTIVATING OBSERVATION

We next test how conventional model merging extends to the low-rank compressed models introduced above. For each layer  $l$ , prior work has considered two natural ways to adapt standard merging to low-rank factors (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025):

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l B_i^l\}_{i=1}^n),$$

where each low-rank update is first reconstructed into full-rank form and then merged.

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{A_i^l\}_{i=1}^n) \mathcal{M}(\{B_i^l\}_{i=1}^n),$$

where the low-rank factors are merged independently and then recomposed.

Both approaches are valid extensions, but they have very different system implications. The combined strategy stores the merged update as an  $m \times d$  matrix, whereas the separate strategy retains the low-rank factor structure through  $A_i^l \in \mathbb{R}^{m \times r}$  and  $B_i^l \in \mathbb{R}^{r \times d}$ . Since  $r(m + d) \ll md$  for  $r \ll \min(m, d)$ , storing a combined full update removes the compression benefit that motivates the setting. For example, with  $m = d = 1024$  and  $r = 16$ , the combined approach stores  $md = 1,048,576$  parameters, while the separate approach stores  $r(m + d) = 16 \times (1024 + 1024) = 32,768$  parameters.

This corresponds to a **32 $\times$  reduction in storage**, so the combined strategy undermines the efficiency benefits of low-rank compression in realistic deployments. We therefore use the separate strategy as the practical baseline for merging compressed models in the remainder of the study.

Using this separate strategy, we apply several existing merging baselines to low-rank compressed models over multiple compression ratios. The motivating experiment uses RoBERTa-base (Zhuang

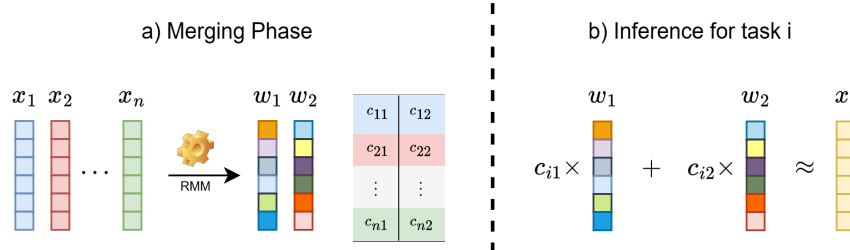


Figure 3: Overview of the RMM framework for reversible low-rank merging. **Left:** During the merging phase, RMM computes a compact set of basis directions (e.g., two in this example) and task-specific coefficients. **Right:** At inference, target model  $i$  is reconstructed by linearly combining the basis vectors with the corresponding coefficients.

et al., 2021) on 5 GLUE tasks (Wang et al., 2018), with details in Appendix A.3. Figures 1 and 2 show that, in this evaluated setting, merging compressed models causes a *catastrophic drop* in average task performance relative to merging full-rank models. We attribute this degradation to two factors: (i) **Amplified task interference:** compression removes representational capacity, making low-rank models more susceptible to destructive interference during merging; (ii) **Loss of coupling between  $A$  and  $B$ :** conventional merging treats low-rank matrices  $A_i^l$  and  $B_i^l$  as independent factors, ignoring the coupled product structure and inducing structural misalignment in the merged updates. Together, these effects impair merged low-rank models and suggest that traditional single-model merging is not well matched to compressed representations.

This observation motivates a more general merging objective for compressed models: instead of requiring one merged model to recover all task-specific behavior, can we reconstruct the original compressed models, or close approximations, from a compact set of basis models whose size depends on the storage budget? Rather than forcing all updates into a single representation, we explore an alternative design space in which merging constructs a *reconstruction-capable model space*. We next formalize this perspective and introduce *Reversible Model Merging* (RMM).

## 5 REVERSIBLE MODEL MERGING

The empirical failures of conventional merging in the compressed setting point to a broader limitation of single-endpoint merging: one merged model may not faithfully preserve every task-specific behavior represented by low-rank updates. RMM addresses this limitation by changing the target representation. Instead of merging all models into a single set of weights, it constructs a small set of shared components, or basis models, from which the original task models can be approximately reconstructed.

Rather than forcing all  $n$  task-specific models into one merged solution  $\hat{\theta}_{\mathcal{M}}$ , RMM retains a compact set of  $p$  merged components, where  $2 \leq p \ll n$ . These components define a low-dimensional subspace intended to capture the dominant variations across tasks, enabling efficient recovery of each model on demand. The same formulation applies across the low-rank adapters and SVD-compressed updates considered here, with  $p$  controlling the storage cost versus reconstruction accuracy trade-off.

For each layer  $l$ , the relevant objects are the  $r$ -dimensional vectors in the low-rank delta matrices ( $A_i^l$  and  $B_i^l$ ): rows of  $A_i^l$  and columns of  $B_i^l$ , where  $r$  is the compression rank. We treat each row of  $A_i^l$  and each column of  $B_i^l$  as an independent *task vector*, and call a fixed row or column index a *task vector position*. For any fixed position, we collect the corresponding task vectors across all  $n$  models and denote the vector from model  $i$  by  $x_i \in \mathbb{R}^r$ . RMM applies the same operation independently at every such position: it projects the set  $\{x_i\}_{i=1}^n$  into a shared basis so that the vectors can later be reconstructed. Each layer has  $m + d$  task vector positions, consisting of  $m$  rows from  $A_i^l$  and  $d$  columns from  $B_i^l$ , and the algorithm handles each position separately.

At each task vector position, RMM identifies basis vectors  $\{w_1, \dots, w_p\}$ , with  $w_j \in \mathbb{R}^r$  for  $1 \leq j \leq p$ . Their span defines the subspace  $W = [w_1 \dots w_p] \in \mathbb{R}^{r \times p}$  used to approximate every task vector  $x_i$ . We introduce  $C \in \mathbb{R}^{n \times p}$  as the coefficient matrix; its  $i$ -th row  $[c_{i1}, \dots, c_{ip}]$  gives the

scalar weights for reconstructing  $\mathbf{x}_i$  from the basis. The objective is to choose  $\mathbf{W}$  and  $\mathbf{C}$  so that the aggregate reconstruction error over the  $n$  task vectors is minimized:

$$\min_{\mathbf{W}, \mathbf{C}} \sum_{i=1}^n \left\| \mathbf{x}_i - \sum_{j=1}^p \mathbf{c}_{ij} \mathbf{w}_j \right\|_2^2. \quad (3)$$

This representation stores  $rp$  parameters for the basis and  $np$  for the coefficients, for a total of  $p(r+n)$  rather than the  $rn$  parameters needed to keep all task vectors directly. The hyperparameter  $p$  therefore controls how broadly the retained subspace can represent task variation under a fixed storage budget. Figure 3 illustrates the case  $p = 2$ . The optimization in equation 3 can be written in matrix form as:

$$\min_{\mathbf{W}, \mathbf{C}} \left\| \mathbf{X} - \mathbf{C} \mathbf{W}^\top \right\|_F^2, \quad (4)$$

where  $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$  stacks the task vectors at the given position. For any  $\mathbf{W}$ , the optimal  $\mathbf{C}$  is given by the least-squares solution  $\mathbf{C}^* = \mathbf{X} \mathbf{W} (\mathbf{W}^\top \mathbf{W})^{-1}$ . With an orthonormal basis ( $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p, p < r$ ), the problem reduces to:

$$\min_{\mathbf{W}} \left\| \mathbf{X} - \mathbf{X} \mathbf{W} \mathbf{W}^\top \right\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_p. \quad (5)$$

In this form, the stored quantities are  $\mathbf{W} \in \mathbb{R}^{r \times p}$  and  $\mathbf{X} \mathbf{W} \in \mathbb{R}^{n \times p}$ , where  $\mathbf{X} \mathbf{W}$  corresponds to  $\mathbf{C}$  in equation 4.

**Theorem 1** (Optimal Solution to equation 5). *Let  $\mathbf{X} \in \mathbb{R}^{n \times r}$  be zero-centered and  $p < \min\{r, n\}$ . The optimal basis  $\mathbf{W}^*$  that minimizes equation 5 is given by the top- $p$  eigenvectors of the sample covariance matrix  $\mathbf{X}^\top \mathbf{X}$ . Equivalently,  $\mathbf{W}^*$  consists of the top- $p$  right singular vectors of  $\mathbf{X}$ .*

The proof of Theorem 1 is provided in Appendix A.1. The resulting procedure has a merging phase that stores the compact representation and an inference phase that reconstructs the requested task model.

## 5.1 MERGING PHASE

The merging phase constructs a compact representation of the task-specific models through a shared basis. For each layer and each task vector position, namely a fixed row in  $\mathbf{A}_i^l$  or column in  $\mathbf{B}_i^l$ , we gather the vectors across all  $n$  models as  $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ . We compute the mean vector  $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$  and zero-center  $\mathbf{X}$  (i.e.,  $\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$ ). We then perform SVD on the centered matrix  $\mathbf{X} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$  and select the top- $p$  right singular vectors as the optimal orthonormal basis  $\mathbf{W}^*$  minimizing the reconstruction error in equation 5, i.e.,  $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$ . Finally, we compute  $\mathbf{C}^* = \mathbf{X} \mathbf{W}^* \in \mathbb{R}^{n \times p}$ , whose  $i$ -th row  $[c_{i1}, \dots, c_{ip}]$  stores the coefficients for reconstructing  $\mathbf{x}_i$ .

After this offline phase, the original task vectors  $\{\mathbf{x}_i\}_{i=1}^n$  can be discarded, and each task vector position is represented only by  $\mathbf{W}^*$ ,  $\mathbf{C}^*$ , and  $\boldsymbol{\mu}$ . The storage requirement drops from  $rn$  parameters for all task vectors to  $p(r+n)+r$  parameters per task vector position, while retaining the information needed for task-specific reconstruction.

## 5.2 INFERENCE PHASE

At inference time, we follow the common model-merging assumption that the target task for an input is either specified by the user or supplied by an oracle router that selects the appropriate model index. Given target task index  $i$ , RMM reconstructs the corresponding task vector  $\mathbf{x}_i$ , denoted  $\hat{\mathbf{x}}_i$ , from the stored compact representation  $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$  at each task vector position. The matrices  $\mathbf{A}_i^l$  and  $\mathbf{B}_i^l$  are then recovered by concatenating the reconstructed  $\hat{\mathbf{x}}_i$  vectors across the relevant positions, i.e., the rows of  $\mathbf{A}_i^l$  and columns of  $\mathbf{B}_i^l$ .

For each task vector position,  $\mathbf{W}^* \in \mathbb{R}^{r \times p}$  stores the orthonormal basis vectors,  $\mathbf{C}^* \in \mathbb{R}^{n \times p}$  stores the task-specific coefficients, and  $\boldsymbol{\mu} \in \mathbb{R}^r$  stores the mean vector across tasks. The reconstruction linearly combines the basis directions with the coefficients for task  $i$  and shifts the result by  $\boldsymbol{\mu}$ :

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p \mathbf{c}_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The vector  $\hat{x}_i \in \mathbb{R}^r$  approximates the original task vector  $x_i$  that was discarded after merging. Applying this reconstruction independently to every task vector position and layer reconstructs the low-rank delta for task-specific model  $i$ , denoted  $\hat{\Delta}_i$ . Algorithm 1 summarizes both phases, and the next section evaluates whether this reversible representation preserves task performance under the same benchmark conditions used for the baselines.

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**Algorithm 1** Reversible Model Merging (RMM)

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1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 

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## 6 EXPERIMENTS

We evaluate RMM across the compression settings, tasks, and model architectures used in this study. The experiments test whether a reversible low-rank representation can preserve task-specific performance while reducing storage, and how this trade-off compares with conventional single-model merging baselines.

### 6.1 EXPERIMENTAL SETUP

The main evaluation uses eight RoBERTa-base models fine-tuned on eight tasks (Zhuang et al., 2021). Each model is compressed independently using PT-SVD or LoRA at target ranks  $r \in \{16, 32, 64, 128\}$ . We then apply RMM to these low-rank models and compare against Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). The performance of the individual compressed models before merging is reported as an upper reference point. Appendix A.2 broadens the empirical coverage to OPT-1.3b (Zhang et al., 2022) and ViT-B/32 (Radford et al., 2021) for vision tasks, with different numbers of tasks.

Following the GLUE benchmark protocol (Wang et al., 2018), each task is evaluated with its standard metric. QNLI (question-answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) use accuracy. CoLA (linguistic acceptability) uses Matthews correlation, and STS-B (semantic textual similarity) uses Pearson correlation. We report the average across tasks as an aggregate performance measure. Storage is measured as the fraction of param-

| $r$ | Method          | Storage | QNLI  | MRPC  | SST-2 | MNLI  | QQP   | RTE   | COLA  | STS-B | Average |
|-----|-----------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|---------|
| 16  | No merging      | 100%    | 92.02 | 85.68 | 93.69 | 85.21 | 87.19 | 77.98 | 58.05 | 90.59 | 83.80   |
|     | TA              | 13%     | 50.59 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.19  | 31.66   |
|     | TIES            | 13%     | 50.56 | 34.78 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.75  | 31.76   |
|     | DARE            | 13%     | 50.80 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 5.07  | 31.80   |
|     | RMM ( $p = 2$ ) | 50%     | 51.20 | 78.20 | 88.30 | 66.74 | 86.02 | 47.29 | 6.56  | 33.59 | 57.24   |
|     | RMM ( $p = 3$ ) | 69%     | 88.83 | 86.26 | 92.89 | 79.58 | 87.18 | 48.74 | 33.85 | 60.46 | 72.22   |
| 32  | No merging      | 100%    | 92.31 | 85.86 | 94.15 | 86.38 | 88.62 | 78.34 | 59.33 | 90.68 | 84.46   |
|     | TA              | 13%     | 50.59 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.21  | 31.66   |
|     | TIES            | 13%     | 50.56 | 35.19 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.87  | 31.83   |
|     | DARE            | 13%     | 50.72 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.48  | 31.59   |
|     | RMM ( $p = 2$ ) | 44%     | 50.54 | 74.32 | 55.05 | 62.68 | 87.07 | 47.29 | 0.00  | 23.18 | 50.02   |
|     | RMM ( $p = 3$ ) | 59%     | 84.66 | 86.32 | 92.89 | 79.70 | 88.05 | 47.29 | 23.57 | 44.54 | 68.38   |
| 64  | No merging      | 100%    | 92.51 | 86.03 | 94.38 | 87.18 | 90.15 | 78.70 | 61.11 | 90.72 | 85.10   |
|     | TA              | 13%     | 50.61 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.23  | 31.67   |
|     | TIES            | 13%     | 50.58 | 35.13 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.80  | 31.82   |
|     | DARE            | 13%     | 50.58 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 5.22  | 31.79   |
|     | RMM ( $p = 2$ ) | 41%     | 50.53 | 73.10 | 49.08 | 58.48 | 86.98 | 47.29 | 0.00  | 17.50 | 47.87   |
|     | RMM ( $p = 3$ ) | 55%     | 74.14 | 86.20 | 92.43 | 78.83 | 89.00 | 47.29 | 13.16 | 33.70 | 64.34   |
| 128 | No merging      | 100%    | 92.68 | 86.14 | 94.50 | 87.33 | 91.05 | 79.78 | 60.16 | 90.74 | 85.30   |
|     | TA              | 13%     | 50.59 | 33.51 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 4.23  | 31.67   |
|     | TIES            | 13%     | 50.58 | 35.54 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 3.93  | 31.88   |
|     | DARE            | 13%     | 50.61 | 34.78 | 49.08 | 31.82 | 36.82 | 47.29 | 0.00  | 2.41  | 31.60   |
|     | RMM ( $p = 2$ ) | 39%     | 50.54 | 70.78 | 49.08 | 54.69 | 84.68 | 47.29 | 0.00  | 14.53 | 46.45   |
|     | RMM ( $p = 3$ ) | 52%     | 59.20 | 85.80 | 92.32 | 77.50 | 89.02 | 47.29 | 6.56  | 26.94 | 60.58   |

Table 1: GLUE performance when merging eight RoBERTa-base task models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ .

| $r$ | Method          | Storage | QNLI  | MRPC  | SST-2 | MNLI  | QQP   | RTE   | COLA  | STS-B | Average |
|-----|-----------------|---------|-------|-------|-------|-------|-------|-------|-------|-------|---------|
| 16  | No merging      | 100%    | 61.38 | 83.48 | 94.50 | 83.90 | 87.56 | 68.23 | 53.92 | 88.71 | 77.71   |
|     | TA              | 13%     | 50.59 | 33.51 | 72.13 | 34.79 | 64.46 | 47.29 | 0.00  | 21.81 | 40.57   |
|     | TIES            | 13%     | 50.63 | 33.68 | 74.31 | 35.64 | 67.45 | 47.29 | 0.00  | 26.38 | 41.92   |
|     | DARE            | 13%     | 50.63 | 33.51 | 73.74 | 35.97 | 65.35 | 47.29 | 0.00  | 21.62 | 41.01   |
|     | RMM ( $p = 2$ ) | 50%     | 51.13 | 34.14 | 90.71 | 60.48 | 84.41 | 47.29 | 0.00  | 43.14 | 51.41   |
|     | RMM ( $p = 3$ ) | 69%     | 55.92 | 38.14 | 93.12 | 82.95 | 87.30 | 47.29 | 6.56  | 57.81 | 58.64   |
| 32  | No merging      | 100%    | 60.13 | 81.28 | 93.81 | 84.46 | 88.18 | 62.82 | 53.12 | 88.53 | 76.54   |
|     | TA              | 13%     | 50.91 | 68.06 | 54.70 | 33.77 | 65.50 | 55.96 | 0.00  | -0.48 | 41.05   |
|     | TIES            | 13%     | 51.03 | 68.06 | 56.77 | 34.24 | 66.27 | 58.12 | 0.00  | 1.02  | 41.94   |
|     | DARE            | 13%     | 50.96 | 64.64 | 63.07 | 35.12 | 63.66 | 55.60 | 0.00  | 1.01  | 41.76   |
|     | RMM ( $p = 2$ ) | 44%     | 52.08 | 65.39 | 81.54 | 62.46 | 77.06 | 57.40 | 0.00  | 9.95  | 50.73   |
|     | RMM ( $p = 3$ ) | 59%     | 54.62 | 69.97 | 92.20 | 81.29 | 85.97 | 57.40 | 4.64  | 24.08 | 58.77   |
| 64  | No merging      | 100%    | 59.62 | 84.12 | 94.15 | 85.20 | 88.75 | 67.15 | 58.31 | 88.70 | 78.25   |
|     | TA              | 13%     | 54.75 | 68.52 | 77.29 | 37.24 | 36.82 | 47.29 | 0.00  | -3.53 | 39.80   |
|     | TIES            | 13%     | 54.82 | 68.75 | 80.73 | 37.67 | 36.82 | 47.29 | 0.00  | -2.48 | 40.45   |
|     | DARE            | 13%     | 53.71 | 68.58 | 78.56 | 38.33 | 36.82 | 47.29 | 0.00  | -4.28 | 39.88   |
|     | RMM ( $p = 2$ ) | 41%     | 55.74 | 69.28 | 87.27 | 42.81 | 77.72 | 47.29 | 0.00  | 4.91  | 48.13   |
|     | RMM ( $p = 3$ ) | 55%     | 59.07 | 70.14 | 90.83 | 79.11 | 82.06 | 47.29 | 0.00  | 14.70 | 55.40   |
| 128 | No merging      | 100%    | 61.58 | 85.04 | 93.58 | 85.71 | 89.68 | 57.04 | 59.35 | 89.68 | 77.71   |
|     | TA              | 13%     | 51.49 | 69.22 | 54.47 | 37.16 | 63.27 | 47.29 | 0.00  | -1.74 | 40.15   |
|     | TIES            | 13%     | 51.91 | 68.93 | 57.34 | 37.81 | 63.36 | 47.29 | 0.00  | -0.25 | 40.80   |
|     | DARE            | 13%     | 51.97 | 68.99 | 58.60 | 35.64 | 63.33 | 47.29 | 0.00  | -0.44 | 40.67   |
|     | RMM ( $p = 2$ ) | 39%     | 57.18 | 67.83 | 88.53 | 49.54 | 84.64 | 47.29 | 0.00  | 28.00 | 52.88   |
|     | RMM ( $p = 3$ ) | 52%     | 59.13 | 39.19 | 92.09 | 80.08 | 86.39 | 47.29 | 15.45 | 71.18 | 61.35   |

Table 2: GLUE performance when merging eight RoBERTa-base task models compressed with LoRA at ranks  $r = 16, 32, 64, 128$ .

ters retained relative to storing all individual task-specific adapters. Single-merged-model baselines have fixed relative cost  $\frac{r}{rn} = \frac{1}{n}$ , while RMM with  $p$  basis components has cost  $\frac{p(r+n)+r}{rn}$ , as derived in Section 5.1.

## 6.2 RESULTS AND DISCUSSION

Tables 1 and 2 compare RMM with the baselines under PT-SVD and LoRA compression. Across both compression mechanisms and all target ranks  $r \in \{16, 32, 64, 128\}$ , RMM outperforms TA, TIES, and DARE by a large margin in this benchmark setting. At lower ranks ( $r = 16, 32$ ), the

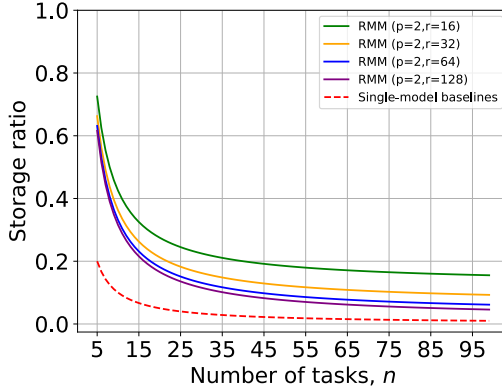


Figure 4: Storage scaling of RMM with  $p = 2$  as the number of models grows. The ratio is measured relative to storing all models.

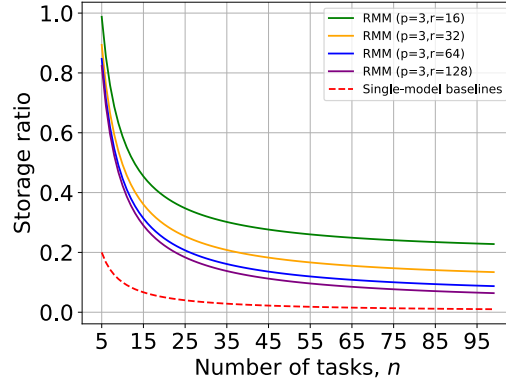


Figure 5: Storage scaling of RMM with  $p = 3$  as the number of models grows. The ratio is measured relative to storing all models.

single-model baselines achieve only 31–42% average score and are especially weak on structurally sensitive tasks such as CoLA and STS-B. RMM improves substantially: at  $r = 16$ ,  $p = 2$  reaches 57.24% under PT-SVD and 51.41% under LoRA, while  $p = 3$  reaches 72.22% and 58.64%, reducing the gap to the individual compressed models. At higher ranks ( $r = 64, 128$ ), RMM remains ahead of the baselines. With  $p = 3$ , it obtains 64.34% and 60.58% under PT-SVD and 55.40% and 61.35% under LoRA, whereas the single-model baselines remain near 40%. Even with  $p = 2$ , RMM maintains clear advantages, supporting the view that reversible reconstruction mitigates the destructive interference seen in prior methods.

The relative gain of RMM over the baselines becomes smaller as rank increases. A plausible explanation is that higher-rank task representations occupy richer and potentially less aligned subspaces, making them harder to summarize with a small shared basis. Overall, the results show that RMM provides a consistent advantage across the evaluated compression regimes and ranks: it preserves stronger task performance while using additional storage that can be tuned through the number of reversible components. This storage is **offline memory**, typically CPU-side and comparatively inexpensive, whereas task performance affects runtime quality. In applications where accuracy is the limiting factor, this modest storage increase can be a favorable trade-off.

### 6.3 SCALABILITY OF RMM

We further examine the scalability of RMM as the number of models,  $n$ , increases. Figure 4 and 5 illustrate the storage ratio (relative to storing all individual task-specific models) as a function of the number of tasks for RMM with  $p = 2$  and  $p = 3$ , respectively. The results confirm that RMM scales favorably: as the number of tasks grows, the relative storage cost decreases significantly. This trend illustrates the main systems benefit of reversible merging: storage grows sublinearly with the number of task-specific models. The reason is structural: once a basis is fixed, adding another task only requires an additional row in the coefficient matrix. Traditional alternatives either store a single merged model, which can sacrifice task performance, or store every task-specific model, which scales linearly. These findings support the use of RMM in larger multi-task, continual, or federated settings where many compressed models must be managed efficiently while retaining task-specific behavior.

## 7 CONCLUSION

This work introduced a reversible framework for model merging that broadens the target representation from a single merged model to a compact reconstructive space. Instead of forcing all task-specific knowledge into one parameter set, RMM stores a small set of intermediate basis representations—fewer than the original task models but more expressive than one aggregate. Within the low-rank settings evaluated here, this design reconstructs task-specific models accurately while

reducing storage overhead. The method therefore provides a controllable trade-off between performance and storage efficiency for LoRA and PT-SVD compressed models. By representing fine-tuned models through compact bases and reversible mappings, RMM enables deployment choices that can be adapted to application-specific memory budgets and reconstruction-fidelity requirements.

Overall, the framework moves beyond one-shot merging by formalizing consolidation as a structured, reversible process. This perspective broadens the design space for efficient multi-task deployment within compressed model families and suggests future work on performance-aware compression, task routing, and reconstruction strategies.

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## A APPENDIX

### A.1 PROOF OF THEOREM 1

Let  $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$ . By definition of the Frobenius norm,  $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$ , where  $\text{Tr}(\cdot)$  denotes trace of a square matrix. Expanding gives:

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

The matrix product inside the trace expands as:

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Next, we take the trace of the expanded form. Using cyclicity of trace and  $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ :

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

| <b>r</b> | <b>Method</b>   | <b>Storage</b> | <b>QNLI</b> | <b>MRPC</b> | <b>SST-2</b> | <b>MNLI</b> | <b>QQP</b> | <b>BOOLQ</b> | <b>Average</b> |
|----------|-----------------|----------------|-------------|-------------|--------------|-------------|------------|--------------|----------------|
| 16       | No merging      | 100%           | 91.74       | 71.54       | 95.30        | 84.19       | 88.14      | 71.47        | 83.73          |
|          | Task-Arithmetic | 17%            | 50.34       | 37.16       | 58.49        | 34.43       | 63.30      | 41.87        | 47.60          |
|          | TIES            | 17%            | 49.79       | 36.75       | 61.24        | 35.57       | 63.55      | 40.67        | 47.93          |
|          | DARE            | 17%            | 49.95       | 37.68       | 56.08        | 34.51       | 63.39      | 42.78        | 47.40          |
|          | RMM ( $p = 2$ ) | 63%            | 88.12       | 44.00       | 93.58        | 73.06       | 85.63      | 59.69        | 74.01          |
|          | RMM ( $p = 3$ ) | 85%            | 91.58       | 49.04       | 95.64        | 80.31       | 87.52      | 65.50        | 78.27          |
| 32       | No merging      | 100%           | 92.06       | 72.17       | 95.41        | 84.60       | 89.17      | 72.45        | 84.31          |
|          | Task-Arithmetic | 17%            | 50.28       | 37.22       | 58.94        | 34.57       | 63.33      | 41.93        | 47.71          |
|          | TIES            | 17%            | 49.77       | 36.93       | 61.93        | 35.69       | 63.65      | 40.73        | 48.12          |
|          | DARE            | 17%            | 50.39       | 40.87       | 56.19        | 34.96       | 63.59      | 43.33        | 48.22          |
|          | RMM ( $p = 2$ ) | 56%            | 86.66       | 41.91       | 91.74        | 71.75       | 85.68      | 56.39        | 72.36          |
|          | RMM ( $p = 3$ ) | 76%            | 91.85       | 44.81       | 94.84        | 80.21       | 88.43      | 63.52        | 77.28          |
| 64       | No merging      | 100%           | 92.26       | 72.12       | 95.41        | 84.64       | 90.07      | 72.84        | 84.56          |
|          | Task-Arithmetic | 17%            | 50.28       | 37.28       | 59.06        | 34.56       | 63.34      | 42.08        | 47.77          |
|          | TIES            | 17%            | 49.75       | 37.04       | 61.81        | 35.84       | 63.74      | 40.76        | 48.16          |
|          | DARE            | 17%            | 50.16       | 40.52       | 58.83        | 35.59       | 63.35      | 42.54        | 48.50          |
|          | RMM ( $p = 2$ ) | 53%            | 83.87       | 42.72       | 89.45        | 69.76       | 85.70      | 53.43        | 70.82          |
|          | RMM ( $p = 3$ ) | 71%            | 92.07       | 41.97       | 94.50        | 79.70       | 88.73      | 61.96        | 76.49          |
| 128      | No merging      | 100%           | 92.35       | 72.41       | 95.41        | 84.67       | 90.93      | 73.09        | 84.81          |
|          | Task-Arithmetic | 17%            | 50.32       | 37.39       | 58.94        | 34.63       | 63.37      | 42.08        | 47.79          |
|          | TIES            | 17%            | 49.79       | 37.04       | 62.04        | 35.99       | 63.74      | 40.55        | 48.19          |
|          | DARE            | 17%            | 50.03       | 38.14       | 57.91        | 34.72       | 63.31      | 42.11        | 47.70          |
|          | RMM ( $p = 2$ ) | 52%            | 81.38       | 42.78       | 84.75        | 67.66       | 83.22      | 51.99        | 68.63          |
|          | RMM ( $p = 3$ ) | 69%            | 92.28       | 41.80       | 93.69        | 79.42       | 88.76      | 59.69        | 75.94          |

Table 3: Benchmark performance when merging six OPT-1.3b task models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ .

Since  $\text{Tr}(\mathbf{X}^\top \mathbf{X})$  is a constant, minimizing  $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$  is equivalent to maximizing  $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$ . Hence,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), the maximizer is obtained by taking the columns of  $\mathbf{W}$  to be the top- $p$  eigenvectors of  $\mathbf{X}^\top \mathbf{X}$ . This is precisely the principal component analysis (PCA) solution: it finds the  $p$ -dimensional subspace that maximizes the projected variance and, equivalently, minimizes the squared reconstruction error. Moreover, if  $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$  is the SVD of  $\mathbf{X}$ , then  $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$ , so the eigenvectors of  $\mathbf{X}^\top \mathbf{X}$  are precisely the right singular vectors (columns of  $\mathbf{V}$ ). Hence, the PCA solution can also be obtained by taking the top- $p$  right singular vectors of  $\mathbf{X}$ . In practice, this SVD-based approach is numerically more stable and often computationally preferable to directly forming  $\mathbf{X}^\top \mathbf{X}$ , especially when  $r$  is large.

## A.2 ADDITIONAL RESULTS

This section extends the empirical scope to two additional pre-trained models while keeping the same reversible-merging protocol. For **OPT-1.3b**, we evaluate six natural language understanding tasks drawn from GLUE and related benchmarks: QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we evaluate vision classification across seven datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010).

Table 3 reports results for six OPT-1.3b models compressed with PT-SVD. Both RMM variants ( $p = 2$  and  $p = 3$ ) outperform the baselines, with  $p = 3$  narrowing the gap to the original individual models while using substantially fewer parameters. Table 4 reports results for seven ViT-B/32 models. On the vision tasks, the benefit of reversible reconstruction is also pronounced: with  $p = 3$ , RMM achieves an average accuracy less than 3% below the original models while reducing storage to 59% of the cost of retaining all models at  $r = 128$ . These additional model classes and domains support the broader applicability of RMM within the evaluated low-rank merging conditions.

| <b>r</b> | <b>Method</b>          | <b>Storage</b> | <b>MNIST</b> | <b>GTSRB</b> | <b>SVHN</b> | <b>Cars</b> | <b>EuroSAT</b> | <b>DTD</b> | <b>SUN397</b> | <b>Average</b> |
|----------|------------------------|----------------|--------------|--------------|-------------|-------------|----------------|------------|---------------|----------------|
| 16       | No merging             | 100%           | 99.61        | 97.22        | 97.06       | 69.20       | 99.67          | 75.21      | 72.72         | 87.24          |
|          | Task-Arithmetic        | 14%            | 47.19        | 28.29        | 38.64       | 56.27       | 57.52          | 41.38      | 62.92         | 47.46          |
|          | TIES                   | 14%            | 60.14        | 34.52        | 49.03       | 58.55       | 61.70          | 43.67      | 62.87         | 52.93          |
|          | DARE                   | 14%            | 53.80        | 31.00        | 44.96       | 57.60       | 56.93          | 43.99      | 62.42         | 50.10          |
|          | <b>RMM</b> ( $p = 2$ ) | 55%            | 99.16        | 88.52        | 94.89       | 66.61       | 97.44          | 60.64      | 67.86         | 82.16          |
|          | <b>RMM</b> ( $p = 3$ ) | 76%            | 99.58        | 95.65        | 96.54       | 68.30       | 98.93          | 67.18      | 69.50         | 85.10          |
| 32       | No merging             | 100%           | 99.65        | 98.45        | 97.35       | 73.04       | 99.78          | 78.72      | 75.52         | 88.93          |
|          | Task-Arithmetic        | 14%            | 47.90        | 28.31        | 38.68       | 56.24       | 57.52          | 41.38      | 63.00         | 47.58          |
|          | TIES                   | 14%            | 61.25        | 35.28        | 49.95       | 58.80       | 62.41          | 44.04      | 63.05         | 53.54          |
|          | DARE                   | 14%            | 53.05        | 31.04        | 46.34       | 58.54       | 57.74          | 42.87      | 62.87         | 50.35          |
|          | <b>RMM</b> ( $p = 2$ ) | 49%            | 98.98        | 88.23        | 94.29       | 68.32       | 96.81          | 61.91      | 68.36         | 82.41          |
|          | <b>RMM</b> ( $p = 3$ ) | 67%            | 99.50        | 96.53        | 96.68       | 70.90       | 98.93          | 70.43      | 71.23         | 86.31          |
| 64       | No merging             | 100%           | 99.67        | 98.68        | 97.41       | 75.76       | 99.85          | 79.15      | 77.73         | 89.75          |
|          | Task-Arithmetic        | 14%            | 47.99        | 28.30        | 38.71       | 56.21       | 57.59          | 41.38      | 63.22         | 47.63          |
|          | TIES                   | 14%            | 61.83        | 35.93        | 50.28       | 59.08       | 63.07          | 44.15      | 63.17         | 53.93          |
|          | DARE                   | 14%            | 54.78        | 32.79        | 45.05       | 58.23       | 61.30          | 42.87      | 62.77         | 51.11          |
|          | <b>RMM</b> ( $p = 2$ ) | 46%            | 98.46        | 84.43        | 92.83       | 69.36       | 95.59          | 62.39      | 69.77         | 81.83          |
|          | <b>RMM</b> ( $p = 3$ ) | 62%            | 99.45        | 96.14        | 96.33       | 72.84       | 98.56          | 72.07      | 73.85         | 87.03          |
| 128      | No merging             | 100%           | 99.68        | 98.75        | 97.46       | 76.99       | 99.85          | 79.04      | 78.59         | 90.05          |
|          | Task-Arithmetic        | 14%            | 48.00        | 28.31        | 38.73       | 56.21       | 57.63          | 41.44      | 63.35         | 47.67          |
|          | TIES                   | 14%            | 62.08        | 36.36        | 50.48       | 59.23       | 63.41          | 44.68      | 63.25         | 54.21          |
|          | DARE                   | 14%            | 55.16        | 32.07        | 45.47       | 58.55       | 61.85          | 43.09      | 62.85         | 51.29          |
|          | <b>RMM</b> ( $p = 2$ ) | 44%            | 97.22        | 78.56        | 90.63       | 70.22       | 94.00          | 61.49      | 71.03         | 80.45          |
|          | <b>RMM</b> ( $p = 3$ ) | 59%            | 99.28        | 94.59        | 95.95       | 74.24       | 98.19          | 71.86      | 76.02         | 87.16          |

Table 4: Vision classification performance when merging seven ViT-B/32 task models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ .

### A.3 RESULTS FOR 5 TASKS

For the motivating experiments in Section 4, we use RoBERTa-base as the pretrained model and evaluate model merging baselines on five GLUE tasks: QNLI, MRPC, SST-2, MNLI, and QQP. Figures 1 and 2 summarize the corresponding aggregate trends from Section 4. For completeness, Table 5 reports the LoRA fine-tuned setting associated with Figure 1, including the improvements from RMM, and Table 6 reports the PT-SVD compressed setting associated with Figure 2.

| <b>r</b> | <b>Method</b>          | <b>Storage</b> | <b>QNLI</b> | <b>MRPC</b> | <b>SST-2</b> | <b>MNLI</b> | <b>QQP</b> | <b>Average</b> |
|----------|------------------------|----------------|-------------|-------------|--------------|-------------|------------|----------------|
| 16       | No merging             | 100%           | 61.38       | 83.48       | 94.50        | 83.90       | 87.56      | 82.16          |
|          | Task-Arithmetic        | 20%            | 50.61       | 33.51       | 73.85        | 34.23       | 64.58      | 51.36          |
|          | TIES                   | 20%            | 50.65       | 33.51       | 75.57        | 35.83       | 66.97      | 52.51          |
|          | DARE                   | 20%            | 50.61       | 33.51       | 79.36        | 34.70       | 63.40      | 52.32          |
|          | <b>RMM</b> ( $p = 2$ ) | 73%            | 59.64       | 68.64       | 93.00        | 82.68       | 86.89      | 78.17          |
|          |                        |                |             |             |              |             |            |                |
| 32       | No merging             | 100%           | 60.13       | 81.28       | 93.81        | 84.46       | 88.18      | 81.57          |
|          | Task-Arithmetic        | 20%            | 51.18       | 67.30       | 58.83        | 34.33       | 68.56      | 56.04          |
|          | TIES                   | 20%            | 51.36       | 68.17       | 59.52        | 34.76       | 71.46      | 57.05          |
|          | DARE                   | 20%            | 50.83       | 64.87       | 68.12        | 35.54       | 68.93      | 57.66          |
|          | <b>RMM</b> ( $p = 2$ ) | 66%            | 57.72       | 67.54       | 92.55        | 81.80       | 86.13      | 77.15          |
|          |                        |                |             |             |              |             |            |                |
| 64       | No merging             | 100%           | 59.62       | 84.12       | 94.15        | 85.20       | 88.75      | 82.37          |
|          | Task-Arithmetic        | 20%            | 54.74       | 68.93       | 80.28        | 35.70       | 36.82      | 55.29          |
|          | TIES                   | 20%            | 55.26       | 68.93       | 78.78        | 36.76       | 36.82      | 55.31          |
|          | DARE                   | 20%            | 54.71       | 68.81       | 74.54        | 38.72       | 36.82      | 54.72          |
|          | <b>RMM</b> ( $p = 2$ ) | 63%            | 59.82       | 67.48       | 91.51        | 83.21       | 84.41      | 77.29          |
|          |                        |                |             |             |              |             |            |                |
| 128      | No merging             | 100%           | 61.58       | 85.04       | 93.58        | 85.71       | 89.68      | 83.12          |
|          | Task-Arithmetic        | 20%            | 52.52       | 68.70       | 54.82        | 37.83       | 63.46      | 55.47          |
|          | TIES                   | 20%            | 52.55       | 69.04       | 57.80        | 38.91       | 63.72      | 56.40          |
|          | DARE                   | 20%            | 55.04       | 69.62       | 54.47        | 38.92       | 64.12      | 56.43          |
|          | <b>RMM</b> ( $p = 2$ ) | 62%            | 60.21       | 34.96       | 91.86        | 78.59       | 86.65      | 70.45          |
|          |                        |                |             |             |              |             |            |                |

Table 5: GLUE performance when merging five RoBERTa-base task models compressed with LoRA at ranks  $r = 16, 32, 64, 128$ .

| <b>r</b> | <b>Method</b>          | <b>Storage</b> | <b>QNLI</b> | <b>MRPC</b> | <b>SST-2</b> | <b>MNLI</b> | <b>QQP</b> | <b>Average</b> |
|----------|------------------------|----------------|-------------|-------------|--------------|-------------|------------|----------------|
| 16       | No merging             | 100%           | 92.02       | 85.68       | 93.69        | 85.21       | 87.19      | 88.76          |
|          | Task-Arithmetic        | 20%            | 53.03       | 35.19       | 49.08        | 31.82       | 36.82      | 41.19          |
|          | TIES                   | 20%            | 55.46       | 54.32       | 49.08        | 31.82       | 37.21      | 45.58          |
|          | DARE                   | 20%            | 53.69       | 58.67       | 49.08        | 31.82       | 37.08      | 46.07          |
|          | <b>RMM</b> ( $p = 2$ ) | 73%            | 79.72       | 85.16       | 90.83        | 76.58       | 86.09      | 83.68          |
| 32       | No merging             | 100%           | 92.31       | 85.86       | 94.15        | 86.38       | 88.62      | 89.46          |
|          | Task-Arithmetic        | 20%            | 53.23       | 35.59       | 49.08        | 31.82       | 36.82      | 41.31          |
|          | TIES                   | 20%            | 55.78       | 55.19       | 49.08        | 31.82       | 37.62      | 45.90          |
|          | DARE                   | 20%            | 55.89       | 33.74       | 49.20        | 31.82       | 36.85      | 41.50          |
|          | <b>RMM</b> ( $p = 2$ ) | 66%            | 63.87       | 82.78       | 91.28        | 77.12       | 87.05      | 80.42          |
| 64       | No merging             | 100%           | 92.51       | 86.03       | 94.38        | 87.18       | 90.15      | 90.05          |
|          | Task-Arithmetic        | 20%            | 53.49       | 35.88       | 49.08        | 31.82       | 36.82      | 41.42          |
|          | TIES                   | 20%            | 56.14       | 55.94       | 49.08        | 31.82       | 38.08      | 46.21          |
|          | DARE                   | 20%            | 53.51       | 33.68       | 49.08        | 31.82       | 36.82      | 40.98          |
|          | <b>RMM</b> ( $p = 2$ ) | 63%            | 56.12       | 81.22       | 91.06        | 76.74       | 88.07      | 78.64          |
| 128      | No merging             | 100%           | 92.68       | 86.14       | 94.50        | 87.33       | 91.05      | 90.34          |
|          | Task-Arithmetic        | 20%            | 53.60       | 36.06       | 49.08        | 31.82       | 36.82      | 41.48          |
|          | TIES                   | 20%            | 56.40       | 56.35       | 49.08        | 31.82       | 38.61      | 46.45          |
|          | DARE                   | 20%            | 50.52       | 34.96       | 49.08        | 31.82       | 36.92      | 40.66          |
|          | <b>RMM</b> ( $p = 2$ ) | 62%            | 54.11       | 79.19       | 90.60        | 74.49       | 87.46      | 77.17          |

Table 6: GLUE performance when merging five RoBERTa-base task models compressed with PT-SVD at ranks  $r = 16, 32, 64, 128$ .